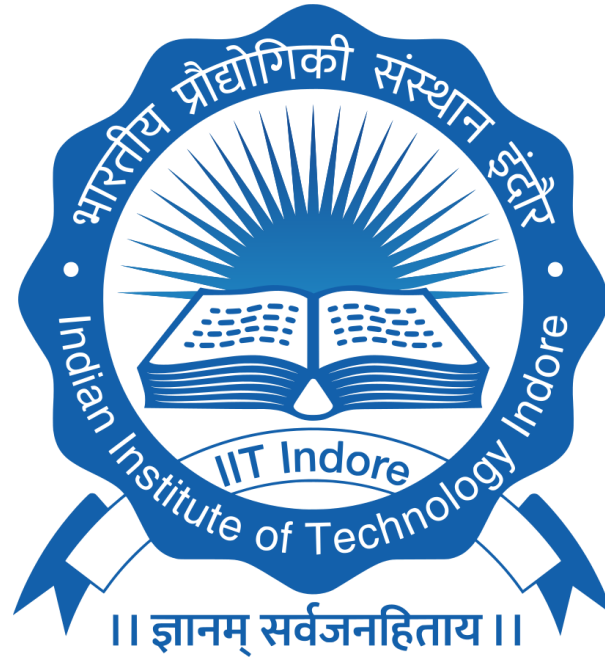


# Mechatronics And Instrumentation Lab

Automatic Centre Stand for Motorcycles



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# ABSTRACT

## **Automatic Centre Stand for Motorcycles**

Parking a motorcycle safely and with minimal effort is important for riders of all age groups. However, traditional centre stands require considerable manual force and proper technique to lift almost half the bike's weight, which becomes difficult for elderly people, women, and riders with physical limitations. As motorcycles become heavier with modern features, the effort needed to use a manual centre-stand increases, often leading to imbalance, strain, or improper parking.

To solve this problem, the project focuses on developing an **Automatic Centre Stand System** that works using a high-torque electromechanical actuator controlled by an Arduino-based circuit. With just a button press, the actuator lowers the stand, lifts the bike smoothly, and locks it in a stable position without any physical lifting. Safety components like limit switches, a motor driver, and current sensing are included to prevent overload, ensure accurate movement, and improve reliability.

This system not only simplifies parking but also improves accessibility and safety for all riders. The report explains the background, motivation, system architecture, mechanical design, electronic circuitry, firmware logic, experimental testing, results, limitations, and future upgrades such as wireless control and IoT-based monitoring. The project aims to offer an engineering-focused yet easy-to-understand solution that enhances everyday motorcycle use.

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# **I. Methodology and System Design**

## **1.1 Introduction**

Motorcycle stand mechanisms have remained largely unchanged for decades, relying on manual levers, pivot joints, and welded steel frames that depend heavily on the rider's physical strength. While these traditional systems are mechanically reliable, they require the rider to lift a significant portion of the motorcycle's weight—often between 40–60% depending on the model. This becomes increasingly challenging for elderly riders, women, individuals with physical limitations, and even average users handling heavier motorcycles such as cruisers, touring bikes, or vehicles carrying luggage. As urban environments grow denser and parking spaces become more restricted, the need for stable and space-efficient parking solutions becomes more critical.

In many real-world scenarios, riders choose to use the side stand simply because deploying the centre stand is physically demanding. However, side stands offer limited stability, pose a higher risk of the motorcycle toppling, and occupy more lateral space in tight parking areas. This issue becomes more pronounced in locations such as college campuses, commercial parking complexes, office premises, and residential apartment lots where motorcycles are parked closely together. A reliable centre stand not only enhances parking safety but also ensures that motorcycles occupy minimal ground space.

The motivation for this project comes from observing the practical difficulties riders face in operating manual centre stands and the safety risks associated with improper parking. This project aims to offer a low-cost, energy-efficient, and universally adaptable automatic centre stand system using an electromechanical actuator and a compact control circuit. By automating the lifting process, the system reduces physical effort, improves stability, and enhances accessibility for a wide range of users. The proposed solution contributes to rider comfort, parking safety, and overall usability, making it suitable for both urban and rural applications.

## **1.2 Literature Review**

Existing automated centre stand solutions mainly replace manual lifting with a linear actuator powered by the motorcycle's 12V battery (we used a 9V battery). These actuators use a DC motor and screw mechanism to generate strong linear motion, lifting the bike upright through the centre stand's pivot. They are simple, dependable, and require no physical effort from the rider.

However, most of these systems depend on manual electrical control using a DPDT switch or similar polarity-reversing setup. The rider must hold the switch during movement, and the actuator stops only when the user releases it. These designs lack automation and safety features—there is no automatic cutoff at end positions, no protection against motor stalling or overheating, and no mechanism to prevent deployment while the bike is moving. Thus, although they reduce physical effort, they fall short in providing intelligent and safe operation for modern riders.

### 1.3 Project Objective

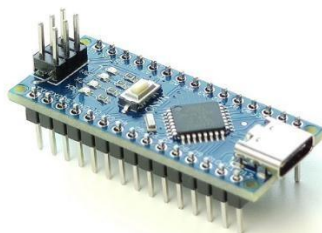
The core objective of this project is to eliminate the substantial physical effort and technique required to lift a motorcycle onto its centre stand, thereby improving accessibility, safety, and overall convenience for riders. This is especially beneficial for elderly individuals, women, and those with physical limitations who often struggle with the weight and balance involved in manually deploying a centre stand. By automating the lifting process, the system also promotes efficient and stable parking in crowded urban environments where precision and ease of use are essential.

To achieve this, the project implements a **automation system** using an Arduino Nano and relay modules to control a high-torque linear actuator. The innovation lies in the addition of automatic motor cutoff using limit switches and the possibility of integrating software-based safety interlocks to prevent misuse or unintended operation. This transforms the traditionally difficult, manual procedure into a smooth, reliable, one-touch operation, significantly enhancing both user safety and the technological value of the motorcycle's parking mechanism.

## II Methodology and System Design

### 2.1 SYSTEM DESIGN AND IMPLEMENTATION

The mechanical section of the system is built around a modified centre stand equipped with an optimized linkage mechanism designed specifically for actuator-driven motion. The stand structure is fabricated from high-strength material capable of bearing the full load of the motorcycle, while pivot joints secure it firmly to the chassis. A linear actuator mechanism—made from a TT gear motor, an M6 threaded rod, and a sliding nut housed inside a protective tube—converts the motor's rotation into smooth linear movement. As the rod extends, it pushes the stand downward, allowing it to make contact with the ground and gradually lift the motorcycle to a stable upright position without requiring manual effort.



The electrical system uses an **Arduino Nano** as the central controller, which receives input from two user-operated buttons corresponding to the extend and retract functions. Based on this input, the Arduino triggers one of two relays arranged in an H-bridge configuration to reverse the motor's polarity and control actuator direction. To ensure safety and prevent mechanical stress, limit switches are installed at both ends of the actuator. These switches send feedback signals when full extension or retraction is reached, allowing the Arduino to cut off power immediately and avoid motor stalling or actuator over-travel. This integrated design ensures smooth, reliable, and protected operation of the automated centre stand.

## 2.2 Mechanical Design & Components

This section explains the mechanical design, fabrication, and assembly of the automated linear actuator used in the system. The primary objective of the mechanical subsystem is to convert the rotary motion of a DC gear motor into smooth and controlled linear motion. The actuator is designed to operate in an automated loop, reversing direction when we press the retract limit switches. The entire mechanical design is organised into three major sub-assemblies:

1. **Drive Unit** – The motor and its supporting structure.
2. **Transmission Assembly** – The mechanism that converts rotary motion into linear displacement.
3. **Mechanical Sensing System** – Push Buttons that physically define the actuator's travel boundaries.

### 2.2.1 Drive Unit

The system is powered by a **TT DC gear motor**, selected specifically for its high torque and low-RPM characteristics. This makes it ideal for lifting loads through a mechanical transmission without requiring a separate gearbox. The motor is rigidly mounted onto the main chassis so that all rotational torque is efficiently transferred into the lead screw mechanism. This stable mounting ensures vibration reduction, alignment stability, and long-term mechanical reliability during repetitive forward–reverse motion.

### 2.2.2 Transmission Assembly (Lead Screw Mechanism)

The transmission assembly forms the core of the mechanical system and uses a **lead screw mechanism** to achieve precise linear movement. The design follows a “rod-in-a-rod” telescoping concept, offering compactness, stability, and smooth actuation.

- **Lead Screw:**  
A 6 mm (M6) steel threaded rod acts as the primary lead screw. It is connected to the motor shaft using a 3D printed motor and thread bar connector, which compensates for minor angular misalignment and prevents binding, vibration, and excessive wear.
- **Actuator Rod (Moving Rod):**  
A hollow tube slides telescopically over the lead screw assembly. An M6 T-nut fixed at its base acts as the captive nut. As the lead screw rotates, this T-nut travels linearly along the threaded rod, pushing or pulling the actuator rod.
- **Cover Rod (Fixed Housing):**  
This hollow outer tube supports the motor and guides the actuator rod along its travel path. It protects internal components and ensures stable, aligned motion.

When the motor rotates (via relays connected to pins D6 and D7), the threaded rod drives the T-nut forward or backward, enabling smooth extension and retraction of the actuator rod.

### 2.2.3 Mechanical Sensing System

Proper control of the actuator requires accurate knowledge of its position. This is accomplished using **two mechanical micro-switches** functioning as end-stop push buttons, which define the total stroke length of the actuator.

- **Extend stop Push Button (D4):**  
Mounted at the maximum extension point.  
When the actuator rod reaches this position, it physically presses the button. The Arduino detects this (LOW input) and immediately activates the `stopMotor()` function.
- **Retract stop Push Button (D5):**  
Positioned at the full retraction or “home” position.  
When triggered, the Arduino executes the `stopMotor()` function.

These switches, combined with the lever-type command inputs (D4 and D5), create a **automated mechanical-electronic system**. The actuator cycles back and forth until explicitly stopped, ensuring reliable, safe, and precise operation.

## 2.3 Electronic Hardware and Interfaces

This section describes the electronic components used to build the smart control system and explains how each module interfaces with the Arduino Nano to form a stable, safe, and responsive control loop. The hardware architecture focuses on low-power logic control, highpower motor actuation, and protective measures to ensure reliable operation under real motorcycle environmental conditions such as vibration, voltage fluctuations, and mechanical shocks.

### 2.3.1 Control Unit Selection and Power Regulation

#### **Microcontroller (Arduino Nano R3):**

The Arduino Nano acts as the main control unit due to its compact size, low power use, and enough digital I/O pins for handling push buttons, limit switches, and relay signals. Its ATmega328P chip ensures stable operation, accurate timing, and smooth interfacing with external protection circuits.

#### **Power Interface and Protection:**

The TT motor runs on ~9 V, while the motorcycle battery provides 12 V, so the Arduino requires a regulated 5 V supply. A 5 V voltage regulator or DC–DC buck converter steps down the voltage safely, protecting the microcontroller from fluctuations and ignition noise. Filtering components like capacitors and diodes further reduce electrical interference and guard against reverse polarity or back-EMF from the motor.

### 2.3.2 User Input Interface

#### Limit Switches:

Two tactile limit switches serve as the user control interface:

- One switch initiate **Extend** (Deploy Stand).
- The second switch triggers **Retract** (Lift Stand).

#### Interface Method:

Each switch is wired to a separate digital input pin on the Arduino. To ensure clean logic signals and prevent floating states, the buttons use either external **pull-down resistors** or the Arduino's internal **pull-up resistors** when wired to ground. This guarantees accurate detection of HIGH and LOW states, reducing the risk of false triggering due to electrical noise or vibrations.

### 2.3.3 High-Power Motor Driver (H-Bridge System)

#### Relay Modules (5 V Relays):

The system uses two electromechanical relays that act as high-current switches, allowing the Arduino's low-power control signals to operate the higher-power TT gear motor. The relays isolate the Arduino from the motor circuit, protecting it from current surges and electrical backflow.

#### H-Bridge Configuration:

The two relays are arranged in an **H-Bridge configuration**, enabling the Arduino to reverse the polarity of the voltage applied to the motor terminals. By toggling specific relays ON or OFF, the controller can run the motor in:

- **Forward Direction** → Deploy Centre Stand
- **Reverse Direction** → Retract Centre Stand

This configuration enables full bidirectional control of the actuator without needing a dedicated motor driver IC, making it simple, cost-effective, and suitable for high-torque gear motors.

## III. Control Logic and Implementation (The Novelty Section)

### 3.1. Control Software Development

```
// INPUTS
const int leverForward = 2;
const int leverReverse = 3;
const int stopButtonFwd = 4;
const int stopButtonRev = 5;

// OUTPUTS
const int relayForward = 6;
const int relayReverse = 7;

const int RELAY_ON = LOW; // active LOW
const int RELAY_OFF = HIGH;

enum State { STOPPED, FORWARD, REVERSE };
State motorState = STOPPED;

// For edge detection:
bool lastLeverFwd = HIGH;
bool lastLeverRev = HIGH;

void setup() {
  pinMode(leverForward, INPUT_PULLUP);
  pinMode(leverReverse, INPUT_PULLUP);
  pinMode(stopButtonFwd, INPUT_PULLUP);
  pinMode(stopButtonRev, INPUT_PULLUP);

  pinMode(relayForward, OUTPUT);
  pinMode(relayReverse, OUTPUT);

  stopMotor();
}

void loop() {
  bool leverFwdNow = digitalRead(leverForward);
  bool leverRevNow = digitalRead(leverReverse);

  bool stopFwdPressed = (digitalRead(stopButtonFwd) == LOW);
  bool stopRevPressed = (digitalRead(stopButtonRev) == LOW);

  // 🛑 STOP BUTTONS ALWAYS HAVE PRIORITY
  if (stopFwdPressed || stopRevPressed) {
    stopMotor();
    motorState = STOPPED;
  }

  // 🟢 Detect SINGLE PRESS on forward lever
  if (lastLeverFwd == HIGH && leverFwdNow == LOW) { // falling edge
    motorState = FORWARD;
    startForward();
  }

  // 🟡 Detect SINGLE PRESS on reverse lever
  if (lastLeverRev == HIGH && leverRevNow == LOW) { // falling edge
    motorState = REVERSE;
    startReverse();
  }

  lastLeverFwd = leverFwdNow;
  lastLeverRev = leverRevNow;

  delay(20);
}

// ---- MOTOR CONTROL FUNCTIONS ----
void startForward() {
  digitalWrite(relayReverse, RELAY_OFF);
  delay(20);
  digitalWrite(relayForward, RELAY_ON);
}

void startReverse() {
  digitalWrite(relayForward, RELAY_OFF);
  delay(20);
  digitalWrite(relayReverse, RELAY_ON);
}

void stopMotor() {
  digitalWrite(relayForward, RELAY_OFF);
  digitalWrite(relayReverse, RELAY_OFF);
}
```

### 3.2. Actuation Sequence Logic (Deployment & Retraction)

The actuation sequence is managed entirely by the Arduino Nano's firmware, which acts as a state machine. The primary goal of the logic is twofold: to control the direction of the motor and to ensure automatic, safe shutdown when the stand reaches its final position.

| Step                  | Action               | Description                                                                                                                                                                                                                    |
|-----------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Input Received        | Read User Command    | The Arduino detects a momentary HIGH signal from either the Extend Limit switch or the Retract Limit Switch.                                                                                                                   |
| Initial State Check   | Verify Current State | The system checks the state of the push buttons to confirm the actuator is not already at the requested endpoint. If the user presses "Extend" but the "Extended stop push button" is already pressed, the request is ignored. |
| Activate Relays       | Establish H-Bridge   | The Arduino activates the specific combination of the two relays necessary to send 9V power to the TT Gear Motor with the required polarity (forward for extend, reverse for retract).                                         |
| Movement Period       | Motor Runs           | The linear actuator pushes/pulls the centre stand. The Arduino remains in a waiting loop, continuously monitoring the status of the appropriate push buttons.                                                                  |
| Push Button Interrupt | Trigger Auto-Stop    | The actuator rod physically contacts the relevant push button (e.g., the Extended push button during deployment). The switch sends a signal (typically a HIGH to LOW transition) back to the Arduino.                          |
| Deactivate Relays     | Cut Motor Power      | The Arduino immediately deactivates both relays in the H-Bridge. This cuts the main 9V power supply to the motor and ensures the flyback diodes absorb any residual inductive energy, protecting the components.               |
| Standby               | System Rest          | The system returns to an idle, low-power state, waiting for the next user command.                                                                                                                                             |

#### Core Actuation Sequence Steps

The logic follows a consistent sequence for both deployment (Extend) and retraction (Retract):

### **3.3. Advanced Safety Interlocks (Novelty Highlight)**

The inclusion of the Arduino Nano allows the system to move beyond simple actuation into the domain of intelligent safety control. These interlocks, managed entirely by the microcontroller's software, are the primary point of novelty compared to baseline automated designs, offering significant advantages in reliability and user safety.

#### **Automatic Motor Cutoff via Push Buttons:**

A major weakness of the DPDT switch design is the need for the user to manually release the switch the moment the stand locks into place. Failure to do so causes the actuator motor to stall against its mechanical stop, leading to high-current draw, rapid overheating, and ultimately, premature failure of the motor and a strain on the battery.

#### **Limit Switch Integration for Safety and Motor Protection**

##### **1. Continuous Monitoring**

When the Arduino begins an extend or retract command, the program enters a monitoring loop. During this process, the microcontroller continuously checks the digital input pins connected to the respective push buttons. This real-time monitoring ensures that the system always knows the actuator's exact position.

##### **2. Instantaneous Cutoff**

The moment the moving actuator rod presses a push button, the signal on the input pin changes state. The Arduino immediately responds by switching both relay control pins to LOW, cutting off the 9 V power supply to the motor without delay. This rapid shutdown prevents the actuator from exceeding its mechanical travel limits.

##### **3. Component Protection**

This automatic cutoff mechanism ensures that the motor never stalls after deployment of the central stand. Preventing stall conditions protects the motor from overheating, reduces stress on mechanical components, extends lifespan, and minimises unnecessary current draw from the motorcycle battery.

## 4. ADVANTAGES

The automated centre stand provides multiple user-focused and mechanical benefits. By removing the need for physical lifting, it allows riders of all age groups—including elderly users and those with limited strength—to park their motorcycles with minimal effort. Because the motorcycle stands vertically when supported on the centre stand, the required parking footprint is reduced, making it ideal for congested parking areas. The system also improves overall stability by keeping the motorcycle upright on surfaces where a manual stand might struggle. Additionally, the actuator-driven mechanism makes manual tampering difficult, offering an added layer of anti-theft protection. The design is modular and can be retrofitted onto most motorcycles without permanent modifications.

## 5. LIMITATIONS

Although the automated centre stand system functions effectively, it still has a few practical limitations. The TT gear motor used in the current design offers limited torque, making it suitable mainly for lightweight and medium-weight motorcycles rather than very heavy models. Since the system relies entirely on electrical power, frequent usage adds a small but noticeable load on the motorcycle's battery. Additionally, the actuator assembly is not fully protected against water and dust, which may affect long-term reliability in outdoor environments. The slow operating speed of the motor, while beneficial for control, also results in longer deployment and retraction times compared to commercial actuators.

## 6. FUTURE WORK

Future improvements can significantly enhance the performance and usability of the system. Adding waterproof housings and sealed enclosures will improve durability in harsh weather conditions. Wireless features such as Bluetooth control, mobile app support, or remote key operation can make the stand even more convenient for users. Additional safety enhancements—like ground-level detection sensors, tilt sensors, or motion-lock mechanisms—can prevent accidental deployment in unsafe conditions. Finally, optimising the power circuitry and firmware can reduce energy consumption and extend battery life.

### 1. Variable Speed Control (Soft Start/Stop):

**Action:** Replace the two relays (D6, D7) with a **solid-state L298N H-Bridge Motor Driver**.

**Benefit:** This allows the Arduino to use `analogWrite()` (PWM) signals to control the motor's speed. This would enable a "soft start" and "soft stop" feature, where the motor gently ramps up to speed and slows down, drastically reducing mechanical shock and wear.

### 2. Closed-Loop Positional Control (Servo-Actuator):

**Action:** Add a **rotary encoder** to the motor's shaft or a **potentiometer** geared to the lead screw.

**Benefit:** This gives the Arduino "eyes" to see its *exact* position, speed, and direction at all times. The system could then be programmed with precise commands like "extend 75mm" or "move to 30% of total length."

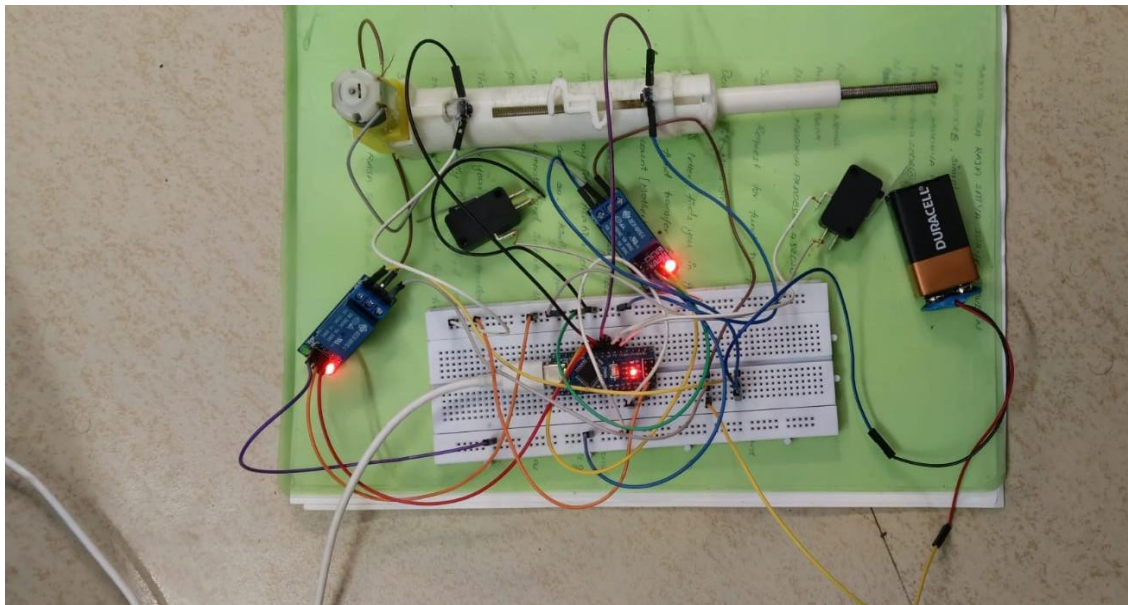
### 3. Overcurrent Protection (Jam Sensing):

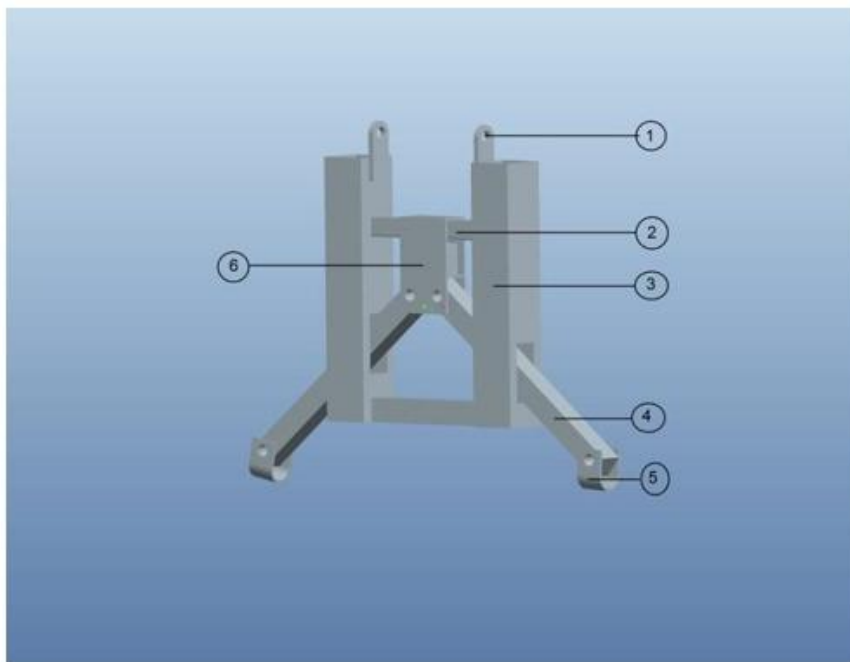
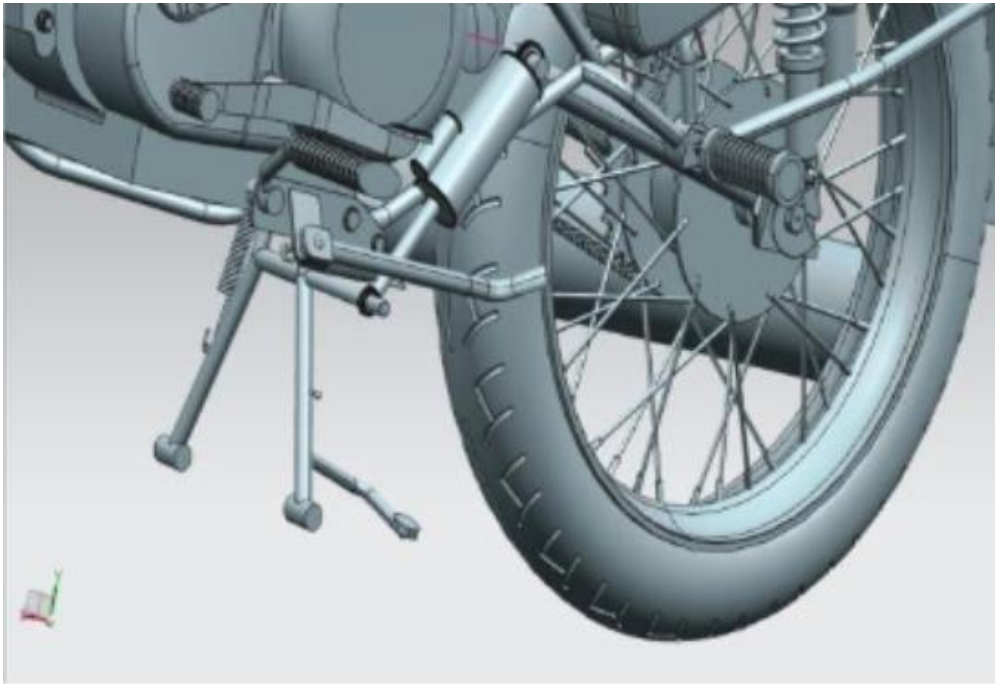
**Action:** Integrate an **ACS712 Current Sensor** on the motor's 12V power line.

**Benefit:** The Arduino can constantly monitor the motor's current draw. If the current spikes above a pre-set limit (a sign of a jam), the program can immediately enter the stopMotor() state, protecting the hardware from damage and alerting a user to a fault.

## 6. CONCLUSION

The automated centre stand system successfully resolves the physical challenges associated with traditional motorcycle parking methods. By integrating mechanical engineering principles with electronic control systems, the project demonstrates how accessible and reliable solutions can be developed using cost effective components. The actuator driven stand eliminates physical effort, enhances safety, and improves stability. With further refinement and stronger actuators, this design can evolve into a commercial grade product capable of benefiting a wide range of motorcycle users.





1. Pivot at vehicle frame
2. Rack and rack for balancing
3. Primary frame
4. Adjustable legs
5. Contact pads
6. Pivot for Actuator